Sri Likithrahulmupparthi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.62
Sri Chaitanya Junior College, Palakollu	Class XII (Board of Intermediate Education Andhra Pradesh)	2021	80.40%
Sri Chaitanya Techno School, Palakollu	Class X (Board of Secondary Education, Andhra Pradesh)	2019	9.20%

INTERNSHIPS

Web Development Intern -- APSCHE & Mayinkrish Ventures Pvt. Ltd. (Mar '25 - Mar '25)

- **Developed** a responsive bank website **using HTML & CSS**, structuring semantic layout and a functional navbar, **improving** page navigability and user flow
- **Engineered** cross-device layout compatibility **through CSS media queries and responsive design techniques, increasing** UI consistency across screen sizes
- **Optimized** overall UI/UX **by restructuring visual hierarchy and page layout, enhancing** usability and reducing user friction

Data & CRM Intern -- International Institute of Digital Technologies (May '24 - Jul '24)

- **Engineered** CRM tool integrations for BlackBuck Engineers **using customer and sales data analysis, improving** lead tracking accuracy and conversion rate
- **Optimized** digital marketing workflows **through campaign performance analysis, driving** higher engagement and campaign efficiency
- **Generated** actionable business insights **using Excel-based data analysis techniques, enabling** data-driven decision-making for stakeholders

Software Engineering Intern (DSA - Java) -- Henotic IT Solutions Pvt. Ltd. (Jul '23 - Sep '23)

- **Implemented** 50+ Data Structures & Algorithms problems **in Java,** applying time/space complexity optimization, **reducing** algorithmic runtime overhead
- **Developed** modular, reusable code **using Object-Oriented Programming principles, improving** code readability and maintainability
- **Strengthened** problem-solving efficiency **by applying complexity analysis to optimize existing algorithms, sharpening** overall code performance

PROJECT

AI-Powered E-Waste Valuation System Using Machine Learning (M.Tech Thesis)

- Built an end-to-end **predictive analytics** solution using **Random Forest Regression** in **Python** (**Pandas**, **NumPy**, **Scikit-learn**) to predict laptop and smartphone resale value, achieving **R^2 0.92**.
- Applied **feature engineering** and **SHAP** to build **interpretable** models that explain value drivers to business users.
- Quantified recoverable value of 5 metals, showing formal recycling (80%) recovers 2x the value of informal (40%), and delivered **business impact** through a 0-100 circular economy score and a **Flask** web app.

Enterprise Retail Analytics & Demand Forecasting Pipeline

- Engineered **RFM** features on 9,994 transactions using **Python** and **Advanced SQL** (**window functions**, **CTEs**, **joins**) for **customer segmentation**.
- Built **time series forecasting** models with **Random Forest** and **Linear Regression** (**supervised learning**), reaching ~83% accuracy and **MAPE** of 17%.
- Identified a loss-making segment despite high revenue and presented **insights and recommendations** in **Power BI** to drive **business decisions**.

Multi-Stage Personalization & Content Recommendation Engine

- Built a **Recommender System** on 100,836 ratings and 9,742 movies using **collaborative filtering** and **SVD matrix factorization**.
- Trained on an **80/20 train-test split** across 610 users, with **model evaluation** showing a 4.4 average consensus rating.
- Analyzed rating patterns using **SQL** in **MySQL** to surface user preference insights and validate recommendation quality.

AI-Powered Smart City Traffic Management

- Built a **Computer Vision** pipeline using **YOLOv8**, **ByteTrack** and **OpenCV** (**Deep Learning**) for **real-time** detection and tracking of vehicles and pedestrians at a 4-lane intersection.
- Trained a **Q-learning Reinforcement Learning** agent for adaptive signal timing, keeping average lane wait under 1.1 seconds in low congestion.
- Detected 11 vehicles and 2 pedestrians in a live test with a flow rate of 13.0 veh/min, giving a **scalable** base for smart city traffic analytics.

Intrusion Detection System Using Machine Learning (B.Tech Final Year Project)

- Benchmarked 6 **classification** models (**Random Forest**, **SVM**, **ANN**, **Gradient Boosted Trees**, **Decision Trees**, Naive Bayes) on NSL-KDD and CICIDS 2017.
- Applied **feature selection** and **feature engineering** using **Scikit-learn** to reduce false positives across 4 attack categories (DoS, Probe, R2L, U2R).
- Selected Random Forest for the best trade-off between accuracy, training time and cost, reaching near 90% accuracy

SKILLS

Languages & Databases: Python, SQL (Joins, CTEs, Window Functions), C, Java, MySQL, PostgreSQL, SQLite

Data & Statistics: Pandas, NumPy, EDA, Data Cleaning, Feature Engineering, Probability, Hypothesis Testing, A/B Testing, ANOVA, PCA

ML, DL & GenAI: Scikit-learn, XGBoost, Regression, Decision Trees, Random Forest, SVM, Clustering, Time Series, Recommender Systems, SHAP, TensorFlow, PyTorch, ANN, Computer Vision (YOLOv8), Reinforcement Learning, LLMs, RAG, LangChain, Prompt Engineering, Agent-based Systems

Visualization & Tools: Power BI, Tableau, Matplotlib, Seaborn, Git, GitHub, Jupyter, Colab, VS Code, Streamlit, FastAPI

Soft Skills: Communication, Analytical Thinking, Problem Solving, Teamwork

POSITIONS OF RESPONSIBILITY

- **Class Representative (CR)**, B.Tech -- SRKR Engineering College -- Acted as liaison between faculty and students, coordinating academic activities and resolving student concerns
- **English Club Leader**, Sri Chaitanya Techno School -- Led club activities and peer sessions, strengthening communication and leadership skills among students

RELEVANT COURSE

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Machine Learning, Statistics & Probability, Data Analytics, Business Analytics, Data Visualization, Operating Systems

CERTIFICATIONS

- **Digital VLSI Design (DSCH Tool)** - Completed, 2022-2023
- **Gemini Certified Student** - Google for Education, 2026

EXTRA-CURRICULARS

- **Solo Performer**, Annual Cultural Fest -- Represented institute on stage, demonstrating discipline and stage presence
- **Active player in Cricket, Basketball, and Chess** -- Demonstrated teamwork and competitive mindset
- **3rd Place, Slow Cycle Race**, IIT Hyderabad